Diversity Reduction in Activation Steering — Abridged

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1 Diversity Reduction in Activation Steering — Short Version

The question. *Activation steering* [**turntrout2023steering**] adds a fixed vector $s \in \mathbb{R}^d$ to a transformer's residual stream at chosen layers to bias behaviour without retraining. Practitioners observe that as $\|s\|$ grows, outputs become not just biased but *narrow* — the same phrasings, the same answer, repeated. We ask why.

The answer in one sentence. Every transformer has normalization layers (RMSNorm / LayerNorm) that project the residual stream onto a sphere of fixed radius. Adding a fixed offset before this projection *must* concentrate the post-normalization distribution toward the pole $\hat{s} = s/\|s\|$; the concentration tightens as $\|s\|$ grows.

1.1 Setup

Let $x \in \mathbb{R}^d$ be a residual-stream activation with $\mu = \mathbb{E}[x]$ and $\Sigma_x = \mathrm{Cov}(x)$. Let s be the steering vector, $m = \mu + s$, $\hat{m} = m/\|m\|$. RMSNorm acts as $v \mapsto v/\|v\|$ up to a learned gain, so we analyze the bare sphere projection

$$g(v) = v/\|v\|, \qquad \phi_s(x) = \widehat{x+s}.$$

The *spherical variance* $V = 1 - \|\mathbb{E}[\phi_s(x)]\| \in [0,1]$ [**mardia2000directional**] is our main summary: $V = 0$ means all outputs project to the same point on the sphere; V close to 1 means they are essentially uniform.

1.2 The Math in Four Lines

One elementary lemma drives everything. For any nonzero u, v,

$$\|\hat{u} - \hat{v}\| \;\le\; \frac{2\,\|u - v\|}{\max(\|u\|, \|v\|)}. \tag{1}$$

Setting $u = x + s,\ v = s$ gives the global **geometric bound**: every post-steering point lies in a spherical cap of radius $2\|x\|/\|s\|$ around the pole $\hat{s}$. Distribution-level bounds follow by expectation:

$$V \;\le\; \frac{2\,\mathbb{E}[\|x\|]}{\|s\|}, \qquad \mathrm{diam}(\phi_s(\mathcal{D})) \;\le\; \frac{4R}{\|s\|}, \tag{2}$$

where $R = \sup \|x\|$. These are exact, distribution-free, and scale as $O(1/\|s\|)$.

Taylor-expanding g around m to first order (with $\delta = x - \mu$) gives the local **covariance bound**:

$$\Sigma_z \;\approx\; \frac{P_{\perp m}\,\Sigma_x\,P_{\perp m}}{\|m\|^2}, \qquad \mathrm{tr}(\Sigma_z) \;=\; \frac{\mathrm{tr}\,\Sigma_x - \hat{m}^\top \Sigma_x \hat{m}}{\|m\|^2}, \tag{3}$$

where $P_{\perp m} = I - \hat{m}\hat{m}^\top$. Two effects: the projector $P_{\perp m}$ annihilates the $\hat{m}$ direction (rank drops by 1 when $\hat{m}^\top \Sigma_x \hat{m} > 0$), and the surviving variation is rescaled by $1/\|m\|^2$ — the strong-steering regime shrinks variance as $O(1/\|s\|^2)$.

The second-order Taylor expansion (sphere curvature) refines the spherical variance itself:

$$V \approx \frac{\operatorname{tr}\Sigma_x - \hat{m}^\top \Sigma_x \hat{m}}{2\,\|m\|^2}, \qquad \|m(s)\|^2 = s^2\|v_{\text{eff}}\|^2 + 2s(v_{\text{eff}} \cdot \mu) + \|\mu\|^2. \tag{4}$$

The denominator is a *quadratic* in s with floor $\|\mu\|^2$, so $V(s)$ on a log–log plot is a smooth curve — flat shelf at $s \ll \|\mu\|/\|v_{\text{eff}}\|$, slope -2 tail at $s \gg \|\mu\|/\|v_{\text{eff}}\|$ — not a straight line. This is the "parabola on log–log" we observe empirically.

One side condition: the bounds correspond to a *reduction* relative to the unsteered baseline only when

$$\|\mu + s\| > \|\mu\| \iff \cos\angle(\mu, s) > -\frac{\|s\|}{2\|\mu\|}. \tag{5}$$

The gate (5) is algebraic; it fails only for steering vectors mildly anti-aligned with the evaluation distribution's residual mean. This matters in practice (see below).

1.3 Experiments

We test on two open-weight models — Qwen2.5-1.5B-Instruct ($d = 1536$) and Meta-Llama-3-8B-Instruct ($d = 4096$), both RMSNorm + RoPE. Input distribution: 1000 documents from `fineweb-edu`, truncated to 256 tokens. Steering vectors: three publicly trained vectors from EasySteer (`happy`, `style` for Qwen; `create` for Llama), plus two random-direction controls — *per-layer-matched* ($\|r_\ell\| = \|s_\ell\|$ per layer) and *aggregate-matched* ($\|\sum_\ell r_\ell\| = \|\sum_\ell s_\ell\|$ across target layers).

19 recording runs total across scales $c \in \{0, 0.5, 1, 2, 4, 8\}$. Each run streams 1000 prompts, intervenes via `nnsight` at target layers, and accumulates a $d \times d$ streaming covariance (hand-rolled Chan–Golub–LeVeque Welford) plus a $K = 1024$ activation reservoir at the final RMSNorm site.

1.4 Headline Findings

1. **The exact geometric bounds always hold.** Claim 3 ($\max \|\phi_s(x) - \hat{s}\| \le 2\|x\|/\|s\|$), Claim 5 (spherical diameter $\le 4R/\|s\|$), Claims 2 and 6 (spherical variance and expected pairwise distance $\le 2\mathbb{E}\|x\|/\|s\|$, $4\mathbb{E}\|x\|/\|s\|$) pass on every run at every scale. These are the pipeline smoke test.

2. **Claim 4 is always inapplicable.** The pairwise Lipschitz bound requires $\|s\| > R$. Both models have $R \approx 10^3$ at the final RMSNorm site while the largest $\|s_{\text{eff}}\|$ we could push through is ≈ 120. The precondition is never satisfied.

3. **Asymptotic $1/\|s\|$ scaling (Claim 8) manifests on Llama, not on Qwen.** Log–log regression slopes for $V(s)$:

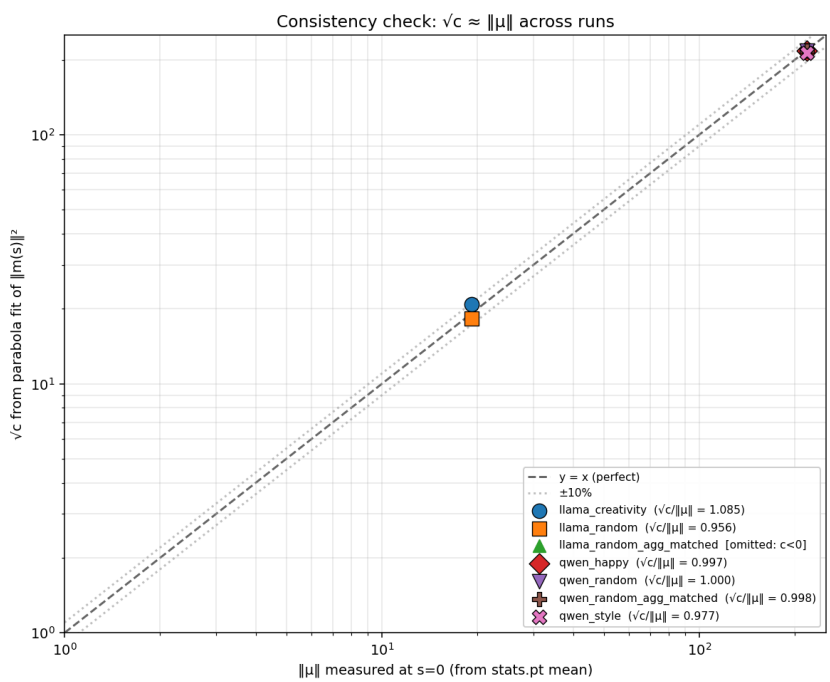
Run	slope V	slope $\operatorname{tr}(\Sigma_z)$
Llama + `creativity` (trained)	**−1.92**	−1.77
Llama + aggregate-matched random	**−1.85**	−1.71
Llama + per-layer-matched random	−0.51	−0.41
Qwen + `style`	−0.29	−0.25
Qwen + aggregate-matched random	−0.20	−0.17
Qwen + `happy`	−0.02	−0.02

Llama at $c = 4, 8$ reaches the regime $\|s_{\text{eff}}\| \gg \|\mu\|$; Qwen never does because Qwen's unsteered $\|\mu\| \approx 218$ is too large to be dominated by any tolerable steering magnitude.

4. **Aggregate magnitude, not semantic direction, drives the scaling.** The trained-vs-aggregate-matched-random gap on Llama is 0.07 on slope V — regression noise. The apparent "trained vectors are special" effect from Experiment 1 alone (trained -1.92, *per-layer*-matched random -0.51) was an artifact of per-layer matching: n near-orthogonal random unit vectors stack to $\sqrt{n}$ in aggregate, whereas coherently trained vectors stack to n. Matching aggregates equates the stacking and erases the gap. **Implication for practice:** random-direction baselines should be aggregate-matched, not per-layer-matched.

5. **The reduction gate (Claim 7) is a precondition, not a theorem.** Qwen + `happy` fails Claim 7 at every scale: $\cos(\mu_{\text{unsteered}}, \sum_\ell s_\ell^{\text{happy}}) = -0.108$ on FineWeb-Edu (educational prose sits slightly on the "sad" side of the happy-vs-sad axis). Single-layer Qwen interventions at layers 17 and 25 fail for *both* real and random directions — a direction-independent, position-dependent effect driven by how little downstream amplification late-layer interventions receive. **Implication for practice:** verify (5) empirically on the evaluation distribution before trusting any of the bounds on a new (model, vector, dataset) triple.

6. **The second-order theory predicts — and the data confirms — the exact shape of $V(s)$.** Fitting $\|m(s)\|^2 = as^2 + bs + c$ on measured $s > 0$ data and extrapolating to $s = 0$ recovers the independently-measured $\|\mu\|$:



Six of seven multi-layer runs land within $\pm 10\%$ of the $y = x$ diagonal. `qwen_random` matches to four decimal places. The log–log parabola of $V(s)$ is not a failure of the slope--1 prediction; it is a theorem: the second-order formula (4) has a quadratic denominator with floor $\|\mu\|^2$, and we can *recover* $\|\mu\|$ from the curvature.

7. **Claim 9 (eigenalignment) is not observed.** The predicted alignment between the smallest eigenvector of Σ_z and $\hat{m}$ does not manifest at the final-RMSNorm site. Most likely cause is that the relevant axis is propagated and rotated relative to the injection direction; future work should evaluate at the injection layer.

1.5 Threats to Validity

- A silent batch-indexing bug contaminated the first run (HuggingFace Transformers ≥ 4.40: Qwen2/Llama3 decoder layers return a bare tensor, not a tuple; `.output[0]` indexes the batch dimension instead of unpacking). All contaminated results are preserved under `discarded/` for audit; all numbers reported above are post-fix.
- `welford-torch` loses $> 100\%$ accuracy in the big-mean / small-batch regime; the pipeline uses a hand-rolled Chan–Golub–LeVeque merge instead. Reproducer filed upstream.
- Qwen + `happy`'s Claim 7 failure is driven by a small but systematic training-pool-vs-evaluation-distribution mismatch. Not a flaw in the theory; a precondition for it to apply.

1.6 What to Take Away

The exact geometric bounds (2) hold unconditionally. The asymptotic scaling law of Claim 8 and the rank-reducing local covariance (3) hold as predicted, but are quantitatively tight only when aggregate steering magnitude exceeds the natural residual-stream scale — a regime Llama-3-8B can reach but Qwen-1.5B cannot within tolerable output quality. The scaling law is a statement about *aggregate magnitude*, not semantic direction; future empirical work should use aggregate-matched random baselines.

The second-order refinement (4) predicts the log–log parabolic shape of $V(s)$ and lets us recover the unsteered $\|\mu\|$ from curvature alone, to within $\pm 10\%$ across every run where the quadratic model is valid.

For full derivations, proofs, per-claim empirical detail, and threats to validity, see the full walkthrough: `docs/paper_sections/diversity_reduction_walkthrough.tex` (29 pages, rendered at `docs/paper_sections/build/diversity_reduction_walkthrough.pdf`).

Code, configs, and per-run outputs are at `https://github.com/AMindToThink/steering_diversity` under `src/bounds/`, `scripts/bounds/`, and `outputs/bounds/`.